littermates (Fig 2, *D*). To assess whether hBD2 directly activates TLR4 in skin-resident cells, we performed live-cell Ca^{2+} imaging on skin-resident cells freshly isolated from mouse ear skin preparations. Consistent with behavioral testing, 10 μM hBD2 elicited a robust $[Ca^{2+}]_i$ response in 4.8% of the skin-resident cells examined (Fig 2, *E*) and the $[Ca^{2+}]_i$ response was completely absent from the skin-resident cells isolated from the *Tlr4*$^{-/-}$ and *LysM*^{cre}; *Tlr4*$^{f/f}$ mice or *Tlr4*$^{-/-}$ DRG neurons (Fig 2, *F-H*). Moreover, hBD2 also activated human skin–resident myeloid cells, which was nearly abolished by a selective TLR4 antagonist LPS-RS (see Fig E4 in this article's Online Repository at www.jacionline.org). Surprisingly, the classic TLR4 ligand LPS from either *Escherichia coli* or *Salmonella enterica* did not evoke significant scratching. Because it is known that itch is constitutively inhibited by pain, we thus tested the effects of intraplantar injections of hBD2 or LPS on mechanical threshold. Indeed, we found that LPS markedly reduced the mechanical threshold while hBD2 had no effect (see Fig E5 in this article's Online Repository at www.jacionline.org), suggesting that the LPS signaling pathway is associated with the pain sensation while the stimulation of TLR4 by hBD2 primarily generates itch sensation without inducing pain responses. Taken together, these results provide strong evidence that TLR4 expressed by skin-resident immune cells but not DRG neurons mediates hBD2-induced itch in mice.

In conclusion, here we first report that hBD2, which is markedly upregulated in differentiated keratinocytes of patients with psoriasis, promotes itch sensation by activating TLR4-expressing cutaneous immune cells in mice. Our findings suggest that hBD2 could act as a potent endogenous pruritogen, which expands the roles of the antimicrobial beta-defensin family and may also provide new therapeutic targets against psoriatic itch.

Jing Feng, PhD[a]
Jialie Luo, PhD[a]
Madison R. Mack, BA[b]
Pu Yang, PhD[a]
Feng Zhang, BS[c]
Guan Wang, BS[c]
Xuan Gong, BS[c]
Tao Cai, MD, PhD[d]
Zhinan Mei, PhD[c]
Brian S. Kim, MD[b]
Shijin Yin, PhD[c]
Hongzhen Hu, PhD[a]

From [a]the Department of Anesthesiology, the Center for the Study of Itch, Washington University School of Medicine, St Louis, Mo; [b]the Division of Dermatology, Washington University School of Medicine, St Louis, Mo; [c]the College of Pharmacy, South-Central University for Nationalities, Wuhan, Hubei, China; and [d]the Department of Dermatology, The First Affiliated Hospital of Chongqing Medical University, Chongqing, China. E-mail: yinshijinyf@163.com. Or: hongzhen.hu@wustl.edu.

This work was supported partly by grants from the National Institutes of Health (grant nos. R01GM101218 and R01DK103901 to H.H.), the Center for the Study of Itch of the Department of Anesthesiology at Washington University School of Medicine (to H.H.), the National Natural Science Foundation of China (grant no. 81373379 to S.Y.), and the Natural Science Fund of Hubei Province (grant no. 2015CFB491 to S.Y.).

Disclosure of potential conflict of interest: B. S. Kim receives grant support from Doris Duke Charitable Foundation, the American Skin Association, and the National Institute of Arthritis and Musculoskeletal and Skin Diseases. S. Yin receives grant support from the National Natural Science Foundation of China and the Natural Science Fund of Hubei Province. H. Hu receives grant support from the National Institutes of Health, the Center for the Study of Itch of the Department of Anesthesiology at the Washington University School of Medicine, and Washington University School of Medicine Digestive Disease Research Core Center (DDRCC). The rest of the authors declare that they have no relevant conflicts of interest.

REFERENCES

1. Yosipovitch G, Goon A, Wee J, Chan YH, Goh CL. The prevalence and clinical characteristics of pruritus among patients with extensive psoriasis. Br J Dermatol 2000;143:969-73.
2. Reich A, Szepietowski JC. Mediators of pruritus in psoriasis. Mediators Inflamm 2007;2007:64727.
3. Jansen PA, Rodijk-Olthuis D, Hollox EJ, Kamsteeg M, Tjabringa GS, de Jongh GJ, et al. β-Defensin-2 protein is a serum biomarker for disease activity in psoriasis and reaches biologically relevant concentrations in lesional skin. PloS One 2009; 4:e4725.
4. Suarez-Carmona M, Hubert P, Delvenne P, Herfs M. Defensins: "Simple" antimicrobial peptides or broad-spectrum molecules? Cytokine Growth Factor Rev 2015;26:361-70.
5. Röhrl J, Yang D, Oppenheim JJ, Hehlgans T. Human β-defensin 2 and 3 and their mouse orthologs induce chemotaxis through interaction with CCR2. J Immunol 2010;184:6688-94.
6. Yang D, Chertov OB, Bykovskaia SN, Chen Q, Buffo MJ, Shogan J, et al. β-defensins: linking innate and adaptive immunity through dendritic and T cell CCR6. Science 1999;286:525-8.
7. Luo J, Feng J, Liu S, Walters ET, Hu H. Molecular and cellular mechanisms that initiate pain and itch. Cell Mol Life Sci 2015;72:3201-23.
8. Riol-Blanco L, Ordovas-Montanes J, Perro M, Naval E, Thiriot A, Alvarez D, et al. Nociceptive sensory neurons drive interleukin-23-mediated psoriasiform skin inflammation. Nature 2014;510:157-61.
9. Biragyn A, Ruffini PA, Leifer CA, Klyushnenkova E, Shakhov A, Chertov O, et al. Toll-like receptor 4-dependent activation of dendritic cells by β-defensin 2. Science 2002;298:1025-9.

Available online April 23, 2017.
http://dx.doi.org/10.1016/j.jaci.2017.03.035

Efficacy and safety of dupilumab for the treatment of adult atopic dermatitis: A meta-analysis of randomized clinical trials

To the Editor:

Dupilumab is a recently commercialized human mAb against the IL-4 receptor α subunit, blocking both IL-4 and IL-13, which have been shown to play important roles in the pathogenesis of atopic dermatitis (AD).[1-3] This study is a meta-analysis evaluating the efficacy and safety of dupilumab for the treatment of AD.

On December 1, 2016, we conducted a systematic search of PubMed, MEDLINE, Embase, and Web of Science databases for randomized controlled trials (RCTs) (published between May 1, 2000, and December 1, 2016; English publication only) using the search terms "dupilumab" and "atopic dermatitis" or "atopic eczema." A total of 7 randomized, double-blinded, placebo-controlled clinical trials (2 phase I, 2 phase IIa, 1 phase IIb, and 2 phase III) including 1965 (1364/601) patients diagnosed with moderate-to-severe AD were selected from 167 publications (see Fig E1 in this article's Online Repository at www.jacionline.org). In the phase I trial M4A, patients received weekly administration of 75, 150, 300 mg dupilumab or placebo for 4 weeks, respectively, and in M4B, patients received weekly doses of 150 or 300 mg dupilumab for 4 weeks. In the phase IIa trial M12, patients received 300 mg dupilumab weekly for 12 weeks,[4] whereas in study C4, patients received a combination treatment of 300 mg dupilumab and topical glucocorticoids for 4 weeks.[4] In the phase IIb study, patients received dupilumab 300 mg once a week (300 mg q1w), 300 mg every 2 weeks (300 mg q2w), 200 mg every 2 weeks (200 mg q2w), 300 mg every 4 weeks (300 mg q4w), 100 mg every 4 weeks (100 mg q4w), or

TABLE I. Characteristics of studies included in the meta-analysis

Reference	Year	Study	Phase	Country	CTG	Treatment/ control, n	Regimen (dose of dupilumab)	Age of dupilumab/placebo group	AD severity of baseline EASI score, treatment/control	Outcome summary
Beck et al[4]	2012	M4A	I	USA	NCT01259323	24/6	75 mg qw, 150 mg qw, 300 mg qw for 4 wk	42.6 ± 1.9/37.4 ± 4.3*	30.0 ± 2.0/22.8 ± 3.0*	Rapid and dose-depedent improvements in all clinical indexes
Beck et al[4]	2013	M4B	I	Multinational	NCT01385657	27/10	150 mg qw, 300 mg qw for 4 wk	42.6 ± 1.9/37.4 ± 4.3*	30.0 ± 2.0/22.8 ± 3.0*	Rapid and dose-dependent improvements in all clinical indexes
Beck et al[4]	2013	M12	IIa	Europe	NCT01548404	55/54	300 mg qw for 12 wk	33.7 ± 1.4/39.4 ± 1.7*	28.4 ± 1.8/30.8 ± 1.9*	Improvement in all clinical indexes in the dupilumab group
Beck et al[4]	2013	C4	IIa	Europe	NCT01639040	21/10	300 mg dupilumab and topic glucocorticoids qw for 4 wk	36.0 ± 2.5/37.8 ± 5.3*	23.1 ± 2.7/24.1 ± 4.0*	Associated with a rapid and sustained reduction in pruritus NRS score
Thaci et al[5]	2014	Phase IIb	IIb	Multinational	NCT01859988	318/61	300 mg qw, 300 mg q2w, 200 mg q2w, 300 mg q4w, 100 mg q4w for 12 wk	37.0 (18.0-75.0)/37.2 (18.0-69.0)†	31.7 ± 0.8/32.9 ± 1.8*	Greater EASI score improvements in the dupilumab group
Simpson et al[7]	2015	SOLO1	III	Multinational	NCT02277743	447/224	300 mg qw for 16 wk, 300 mg q2w for 16 wk	38.0 (27.5-48.0)/39.0 (27.0-50.5)‡	300 mg qw: 30.4 (21.5-40.8), 300 mg q2w: 29.8 (22.0-41.2)/31.8 (22.2-43.8)‡	Clinical index improvement in the dupilumab group
Simpson et al[7]	2016	SOLO2	III	Multinational	NCT02277769	472/236	300 mg qw for 16 wk, 300 mg q2w for 16 wk	34.0 (25.0-46.0)/35.0 (25.0-47.0)‡	300 mg qw: 28.6 (21.0-40.1), 300 mg q2w: 29.0 (21.2-41.8)/30.5 (22.1-41.7)‡	Clinical index improvement in the dupilumab group

CTG, Clinicaltrials.gov identification number; *NRS*, numerical-rating scale.
*Mean ± SE.
†Mean (range).
‡Median (range).

placebo once a week (qw).[5,6] In the 2 latest phase III trials, SOLO1 and SOLO2, patients received 300 mg dupilumab or placebo at either weekly or biweekly basis for 16 weeks.[7] All patients were adults (≥18 years old), had eczema area and severity index (EASI) score of more than 16, investigator's global assessment (IGA) score of more than 3, body surface area affected of more than 10%, and a diagnosis of AD for more than 3 years. The characteristics of the selected studies are presented in Table I. The study qualities of each RCT or subgroup were evaluated using the Jadad scale, and all 7 studies included in the meta-analysis were considered to be of high quality (see Table E1 in this article's Online Repository at www.jacionline.org).[4-7] All 7 trials reported adequate randomization, none was stopped early, and M4B, phase IIb, SOLO1, and SOLO2 trials were conducted in multinational study centers. All 7 studies showed a low risk of bias (see Fig E2, *A*, in this article's Online Repository at www.jacionline.org). Begg funnel plot and Egger test showed that there did not seem to be bias in the publication (Fig E2, *B*).

In the pooled analysis of all 7 trials, significant improvement in efficacy of dupilumab for the treatment of AD was observed in various clinical indexes. Compared with the placebo control, dupilumab treatment significantly reduced the EASI score (standardized mean difference [SMD], −0.91; 95% CI, −0.99 to −0.83; $P < .001$) (Fig 1, *A*), although 3 studies did not show a substantial reduction in the EASI score, including M4, C4, and phase IIb 100 mg q4w group. A higher percentage of IGA response (IGA score turned to 0 or 1) was revealed in dupilumab treatment groups compared with the placebo group (relative risk [RR] = 4.64; 95% CI, 3.81-5.66; $P < .001$) (Fig 1, *B*). In dupilumab treatment groups, the SMD of pruritus numerical-rating scale score was −0.76 (95% CI, −0.84 to −0.68; $P < .001$) (Fig 1, *C*), indicating that dupilumab treatment improved the relief and the life quality of the patients.

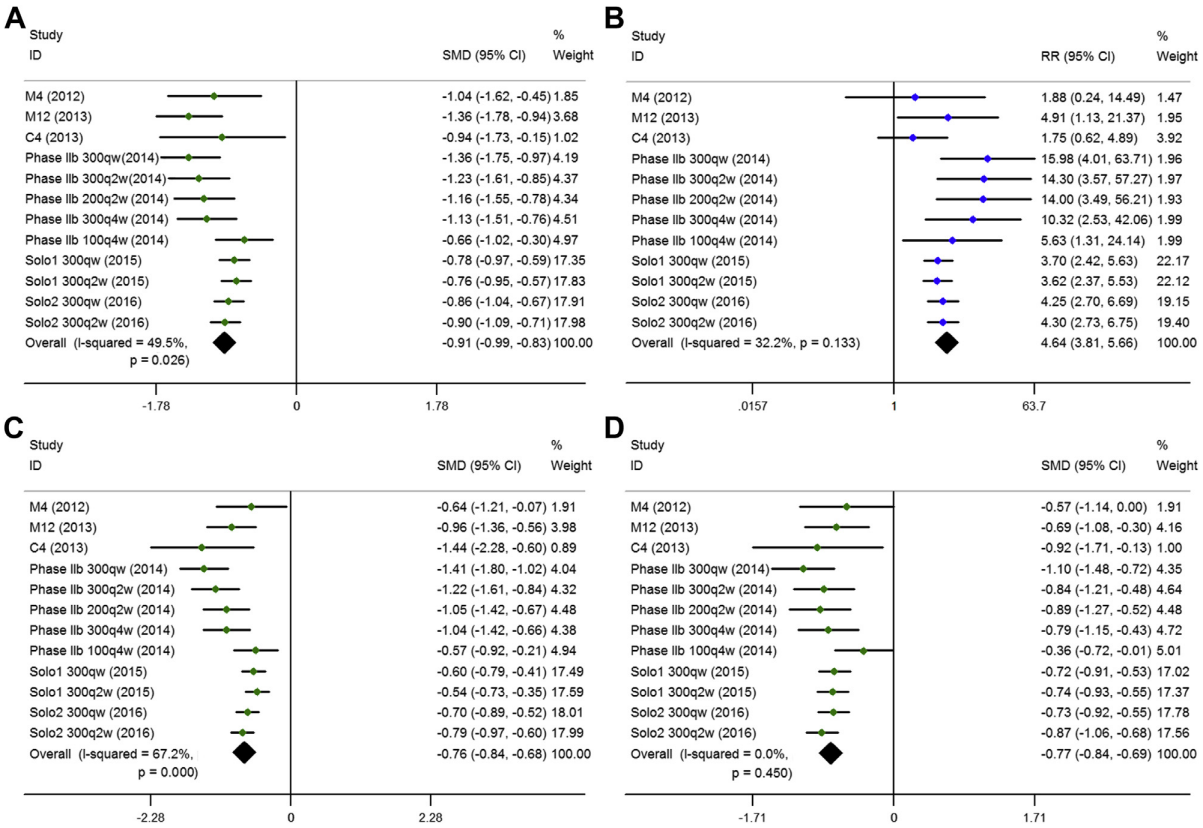

FIG 1. Meta-analysis of efficacy outcomes of 7 RCTs with dupilumab- and placebo-treated patients with AD. **A,** The EASI score. **B,** The IGA score. **C,** Pruritus numerical-rating scale (NRS) score. **D,** Body surface area (BSA) score. *Horizontal lines* stand for 95% CIs of the RR estimates. *Green dots* represent the SMD. *Blue dots* represent the RR, and diamonds represent the meta-analysis summary effect estimate.

The results also showed a significant decrease in the mean difference in body surface area score within the dupilumab groups compared with the control group (SMD, −0.77; 95% CI, −0.84 to −0.69; $P < .001$) (Fig 1, *D*). Furthermore, the dose-dependent efficacy of dupilumab was shown. The results indicated that the most effective clinical improvement was achieved by the doses of 300 mg qw and 300 mg q2w, and the doses of 300 mg q4w and 200 mg q2w also showed significant clinical improvement, but were less effective; meanwhile, the dose of 100 mg q4w was almost ineffective. Moreover, the group receiving 100 mg q4w was the major source of heterogeneity in this meta-analysis (see Fig E3 in this article's Online Repository at www.jacionline.org), confirming that low dose of dupilumab was less likely to achieve satisfactory therapeutic effects as high doses of dupilumab.

Overall, dupilumab was well tolerated and had a favorable safety profile that was substantially better than that of most systemic agents. The commonly reported adverse events (AEs) in patients treated with dupilumab were nasopharyngitis, headache, and injection-site reaction. We found that AEs occurred at a similar rate among different trials without statistical significance (RR = 0.99; 95% CI, 0.95-1.04; $P = .827$) (see Fig E4, *A*, in this article's Online Repository at www.jacionline.org). Dupilumab treatment was associated with lower risk of severe adverse events (SAEs) compared with the placebo group (RR = 0.44; 95% CI, 0.30-0.65; $P < .001$) (Fig E4, *B*). Indeed, the frequency of SAE in the pooled dupilumab group was 2.56%, whereas that of the pooled control group was 6.26%. The major SAEs in the placebo group included skin infection, inflammatory lesion, impetigo, and eczema herpeticum, which possibly resulted from the exacerbation of AD due to the lack of effective treatment. Of note, there was a high incidence (12%) of herpes virus infection in the 100 mg dupilumab group of the phase IIb trial.[5] Finally, similar risks of study discontinuation between dupilumab and placebo groups were observed (RR = 0.95; 95% CI, 0.61-1.50; $P = .839$) (Fig E4, *C*).

To our knowledge, this is the first systematic analysis and comparison of all published dupilumab RCTs data for the treatment of moderate-to-severe AD. The trials included had high consistency in the patients enrolled, randomization and masking, and treatment outcomes. Dupilumab was shown to be consistently more effective than placebo, with a trend of a dose-dependent effect, and had a placebo-like safety profile.[8] Combination of dupilumab with topical glucocorticoids seemed to improve the effects.[9] Limitations of this meta-analysis included the limited number of clinical trials (only 7) in this study and

the risks of bias, for example, funding from pharmaceutical industry, might exist. Our findings suggest that dupilumab is a promising anti-AD medication, due to its specific action directed at the underlying mechanisms of AD, resulting in its good efficacy and safety profile in patients with severe AD. However, further investigations still need to be performed in future studies to evaluate the long-term efficacy and safety of dupilumab for the treatment of AD.

All the research work was conducted in Jiangsu Key Laboratory of Molecular Biology for Skin Diseases and STIs. We thank Xiangdong Gong for help in manipulation of the results.

Yue Han, MD[a]*
Yuxin Chen, PhD[b]*
Xiaochun Liu, MD[a]
Jingxi Zhang, MD[a]
Huichun Su, MD[a]
He Wen, MD[a]
Wei Li, MD, PhD[c]
Xu Yao, MD, PhD[a]

From [a]the Institute of Dermatology, Chinese Academy of Medical Sciences and Peking Union Medical College, Nanjing, [b]the Department of Laboratory Medicine, Nanjing Drum Tower Hospital, Nanjing University Medical School, Nanjing, and [c]the Department of Dermatology, Huashan Hospital, Fudan University, Shanghai, China. E-mail: dryao_xu@126.com. Or: liweiderma@163.com.

*These authors contributed equally to this work.

This work was supported by the National Natural Science Foundation of China (grant nos. 81371735, 81573045, 81573038, and 81600201), the Natural Science Foundation of Jiangsu province (grant no. BK20151066), Open Grant of the Jiangsu provincial Key Laboratory of Molecular Biology for Skin Diseases and STIs (grant no. 2015KF14), and the Milstein Medical Asian American Partnership Foundation.

Disclosure of potential conflict of interest: The authors declare that they have no relevant conflicts of interest.

REFERENCES

1. Suarez-Farinas M, Dhingra N, Gittler J, Shemer A, Cardinale I, de Guzman Strong C, et al. Intrinsic atopic dermatitis shows similar TH2 and higher TH17 immune activation compared with extrinsic atopic dermatitis. J Allergy Clin Immunol 2013;132:361-70.
2. Kaesler S, Volz T, Skabytska Y, Koberle M, Hein U, Chen KM, et al. Toll-like receptor 2 ligands promote chronic atopic dermatitis through IL-4-mediated suppression of IL-10. J Allergy Clin Immunol 2014;134:92-9.
3. Wenzel S, Castro M, Corren J, Maspero J, Wang L, Zhang B, et al. Dupilumab efficacy and safety in adults with uncontrolled persistent asthma despite use of medium-to-high-dose inhaled corticosteroids plus a long-acting beta agonist: a randomised double-blind placebo-controlled pivotal phase 2b dose-ranging trial. Lancet 2016;388:31-4.
4. Beck LA, Thaci D, Hamilton JD, Graham NM, Bieber T, Rocklin R, et al. Dupilumab treatment in adults with moderate-to-severe atopic dermatitis. N Engl J Med 2014;371:130-9.
5. Thaci D, Simpson EL, Beck LA, Bieber T, Blauvelt A, Papp K, et al. Efficacy and safety of dupilumab in adults with moderate-to-severe atopic dermatitis inadequately controlled by topical treatments: a randomised, placebo-controlled, dose-ranging phase 2b trial. Lancet 2016;387:40-52.
6. Simpson EL, Gadkari A, Worm M, Soong W, Blauvelt A, Eckert L, et al. Dupilumab therapy provides clinically meaningful improvement in patient-reported outcomes (PROs): a phase IIb, randomized, placebo-controlled, clinical trial in adult patients with moderate to severe atopic dermatitis (AD). J Am Acad Dermatol 2016;75:506-15.
7. Simpson EL, Bieber T, Guttman-Yassky E, Beck LA, Blauvelt A, Cork MJ, et al. Two phase 3 trials of dupilumab versus placebo in atopic dermatitis. N Engl J Med 2016;375:2335-48.
8. Hamilton JD, Suarez-Farinas M, Dhingra N, Cardinale I, Li X, Kostic A, et al. Dupilumab improves the molecular signature in skin of patients with moderate-to-severe atopic dermatitis. J Allergy Clin Immunol 2014;134:1293-300.
9. Roekevisch E, Spuls PI, Kuester D, Limpens J, Schmitt J. Efficacy and safety of systemic treatments for moderate-to-severe atopic dermatitis: a systematic review. J Allergy Clin Immunol 2014;133:429-38.

Available online May 4, 2017.
http://dx.doi.org/10.1016/j.jaci.2017.04.015

Assessment of thunderstorm-induced asthma using Google Trends

To the Editor:

Google Trends (GT), a Web-based surveillance tool, uses Google to explore the searching trends of specific queries. GT may predict the outbreak of many diseases. In Germany, correlations between the patient-based, combined symptom medication score and GT data are stronger than those with the regionally measured pollen count data.[1] Search activities using GT for terms such as "allergy," "allergies," and "pollen" correlate strongly with observed pollen counts.[2,3] GT reflects the real-world epidemiology of symptomatic allergic rhinitis and could potentially be used to monitor allergic rhinitis.[4] Seasonality of allergic rhinitis was found using Internet searches and correlated with pollen counts.[3] Twitter data, Google search interests, and environmental sensor data can also be used to predict the number of asthma-related emergency department visits in an area.[5]

Studies on thunderstorm-induced asthma have convincingly shown that grass pollen can induce severe asthma exacerbations when climatic conditions and pollen exposure are associated.[6,7] It is possible that such outbreaks can be observed using GT. Two recent outbreaks with several asthma deaths were observed in Melbourne, Australia, November 20, 2016, and in Kuwait, December 2, 2016.

In the present study, we used GT to compare terms related to asthma, allergy, and rhinitis in 10 countries from 2004 to December 20, 2016. The aims of the study were (1) to assess whether GT could report the recent thunderstorm asthma outbreak in Australia and Kuwait and (2) to determine whether such an outbreak could be found in other countries over the past 13 years.

The following terms were used: "rhinitis," "allergic rhinitis," "hay fever," "asthma," "pollen," and "allergy" (disease and term). However, only "allergy," "allergic rhinitis," "asthma," and "pollen" were analyzed because "rhinitis" as a term or "hay fever" were labeled as "allergic rhinitis disease." Only "diseases" and "subjects" are translated by GT, whereas "terms" are not translated. Major antirhinitis (eg, loratadine and cetirizine) or antiasthma (eg, Salbutamol and Seretide) medications were studied. Intranasal and inhaled corticosteroids could not be differentiated using GT and were not used in the study.

We examined GT queries from January 1, 2004 (starting date of GT), to December 20, 2016, for "all countries" and from January 2011 to December 20, 2016, for the 5 most populated European Union countries (ie, France, Germany, Italy, Spain, and the United Kingdom), as well as Australia, Canada, New Zealand, and the United States. Only temperate and continental climatic zone countries were studied because allergen exposure may vary with other climatic zones. We added Kuwait because a severe thunderstorm-induced asthma outbreak occurred on December 2, 2016.

The prediction of the asthma outbreak using "allergic rhinitis," "allergy," "pollen," or "rhinitis medications" queries was not tested because a more detailed analysis should be carried out focusing on the region of the thunderstorm.

METHODS

This meta-analysis was performed and written according to the Preferred Reporting Items for Systematic Reviews and Meta Analyses statement as a guideline (http://www.prisma-statement.org/).

Search strategy and inclusion criteria

On December 1, 2016, we conducted a systematic search of PubMed, MEDLINE, Embase, and Web of Science databases for RCTs (published between May 1, 2000, and December 1, 2016; English publication only) using the search terms "dupilumab" and "atopic dermatitis" or "atopic eczema." A 2-step approach was applied, and all the studies selected using the strategy were reviewed independently by 2 researchers (H.Y. and C.Y.X.); finally, the reports that met all inclusion criteria were included in our meta-analysis. The inclusion criteria were as follows: (1) patients given a diagnosis of AD and treated with dupilumab; (2) double-blind, randomized, placebo-controlled studies; (3) outcome measures including EASI, IGA, and pruritus numerical-rating scale (NRS) score, as well as rates of patients with AEs and the percentage of study discontinuation.

Testing for heterogeneity and publication bias of studies

We performed the Q test (to show statistically significant heterogeneity when $P < .1$) and the I^2 test (measures the proportion of the overall variation that is attributable to between-study heterogeneity [range from 0% to 100%, >50% is considered as evidence of heterogeneity]) to analyze the heterogeneity of the studies included in our meta-analysis. The Galbraith plot was applied to indicate studies that caused heterogeneity. Each study was described as a single dot with a central regression line through the plot. On the y-axis, the log-transformed effect size divided by the SE (z score) was shown, and on the x-axis the inverse of the SE was shown.

Moreover, we analyzed the methodological quality of the included studies and the risk of bias using Cochrane collaboration tool, and we also performed the Egger test, which is a linear regression method, to evaluate the bias in publication in this meta-analysis.

Data extraction and outcome measures

For our meta-analysis, we extracted the following predefined variables: authors, year of publication, study type, name of the study, clinicaltrials.gov number, dupilumab doses used, number of dupilumab-treated patients and control subjects, duration of study periods, and outcome measures. The focus of our analysis was to compare the treatment effects of dupilumab and placebo across studies. The outcomes included in our meta-analysis were EASI, IGA, NRS score, rate of patients with AEs and SAEs, and study discontinuation rate.

Statistical analysis

From studies included in this analysis, we extracted, for dupilumab and placebo, EASI, IGA, NRS score, and the RR for the occurrence of at least 1 AE (along with the corresponding 95% CI). The overall summary effect sizes were estimated using a random-effects model. A P value of less than .05 was considered statistically significant. Review Manager (RevMan version 5.2.5; Nordic Cochrane Centre, Copenhagen, Denmark) and STATA (version 12.0; StataCorp, College Station, Tex) software were used for statistical analysis.

REFERENCES

E1. Beck LA, Thaci D, Hamilton JD, Graham NM, Bieber T, Rocklin R, et al. Dupilumab treatment in adults with moderate-to-severe atopic dermatitis. N Engl J Med 2014;371:130-9.

E2. Thaci D, Simpson EL, Beck LA, Bieber T, Blauvelt A, Papp K, et al. Efficacy and safety of dupilumab in adults with moderate-to-severe atopic dermatitis inadequately controlled by topical treatments: a randomised, placebo-controlled, dose-ranging phase 2b trial. Lancet 2016;387:40-52.

E3. Simpson EL, Gadkari A, Worm M, Soong W, Blauvelt A, Eckert L, et al. Dupilumab therapy provides clinically meaningful improvement in patient-reported outcomes (PROs): a phase IIb, randomized, placebo-controlled, clinical trial in adult patients with moderate to severe atopic dermatitis (AD). J Am Acad Dermatol 2016;75:506-15.

E4. Simpson EL, Bieber T, Guttman-Yassky E, Beck LA, Blauvelt A, Cork MJ, et al. Two phase 3 trials of dupilumab versus placebo in atopic dermatitis. N Engl J Med 2016;375:2335-48.

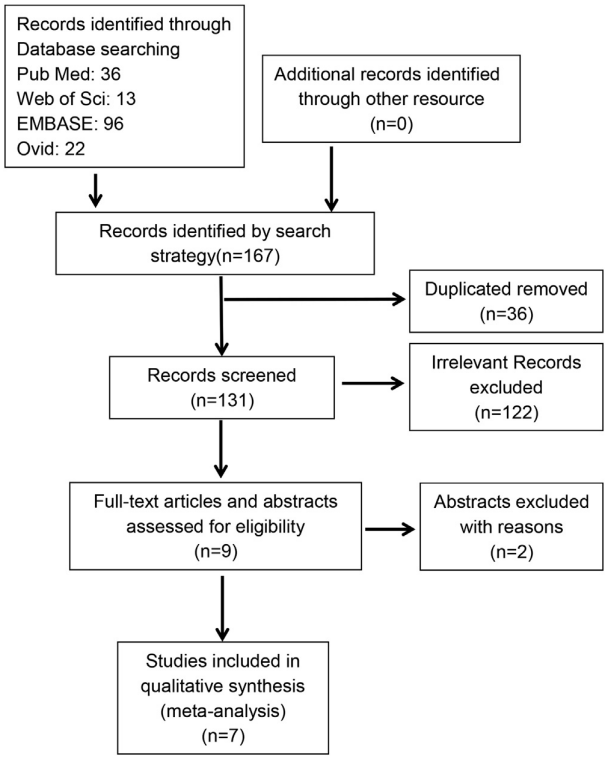

FIG E1. Study selection process. We collected 167 potentially relevant studies, of which 7 were included in the meta-analysis.

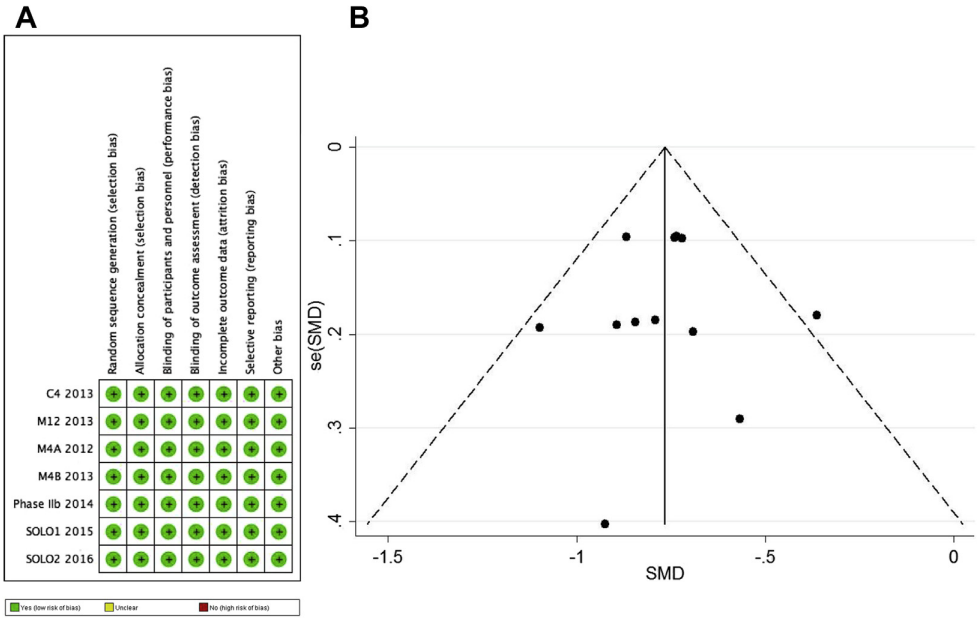

FIG E2. A, Risk of bias summary for each included study. **B,** Begg funnel plot. Studies with larger scale have been spread around the average, and publications with smaller scale were plodded evenly on each side of the average. Deviation from such funnel-like distribution is able to point out the bias of the studies. Both the funnel plot and Egger test showed no evidence of bias in the publication.

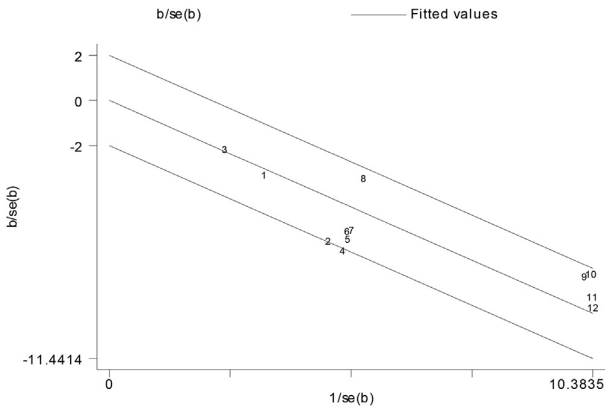

FIG E3. Galbraith radial plot. The figure shows the contribution of results from the patient groups treated with different doses of dupilumab in the different studies to the heterogeneity of efficacy of treatment. The following studies were included: phase IIb 300qw (1), phase IIb 200q2w (2), phase IIb 300q2w (3), phase IIb 100q4w (4), phase IIb 300q4w (5), M4 (6), M12 (7), C4 (8), SOLO1 300q2w (9), SOLO1 300qw (10), SOLO2 300q2w (11), and SOLO2 300qw (12). The group receiving 100 mg every 4 week (100q4w) in phase IIb study was found to be the main cause of the heterogeneity.

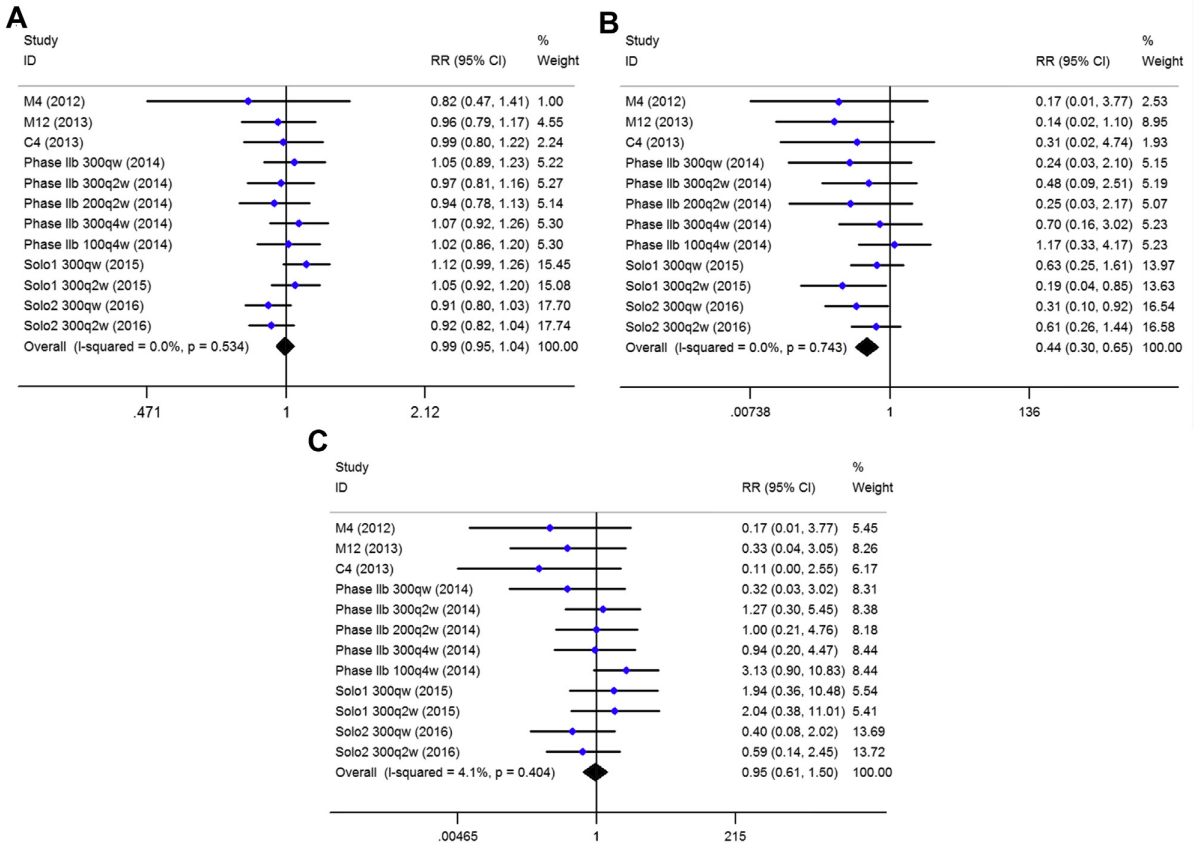

FIG E4. Forest plot of rate of AEs, SAEs, and study discontinuation in 7 RCTs. **A,** AE rates and RR of AE rates between dupilumab- and placebo-treated patients. **B,** SAE rates and RR of SAE rates between dupilumab- and placebo-treated patients. **C,** Rates of study discontinuation and RR of study discontinuation between dupilumab- and placebo-treated patients. *Diamonds* stand for the meta-analysis summary effect estimate. *Blue dots* represent the RR, and *horizontal lines* represent 95% CIs of the RR estimates.

TABLE E1. Quality assessment of included RCTs using the Jadad scale

Reference	Year	Study	Representation of randomization	Appropriateness of method for randomization	Representation of double blinding	Appropriateness of method for double blinding	Representation of withdrawals	Total score
Beck et al[E1]	2012	M4A	1	1	1	1	1	5
Beck et al[E1]	2013	M4B	1	1	1	1	1	5
Beck et al[E1]	2013	M12	1	1	1	1	1	5
Beck et al[E1]	2013	C4	1	1	1	1	1	5
Thaci et al[E2]	2014	Phase IIb	1	1	1	1	1	5
Simpson et al[E4]	2015	SOLO1	1	1	1	1	1	5
Simpson et al[E4]	2016	SOLO2	1	1	1	1	1	5